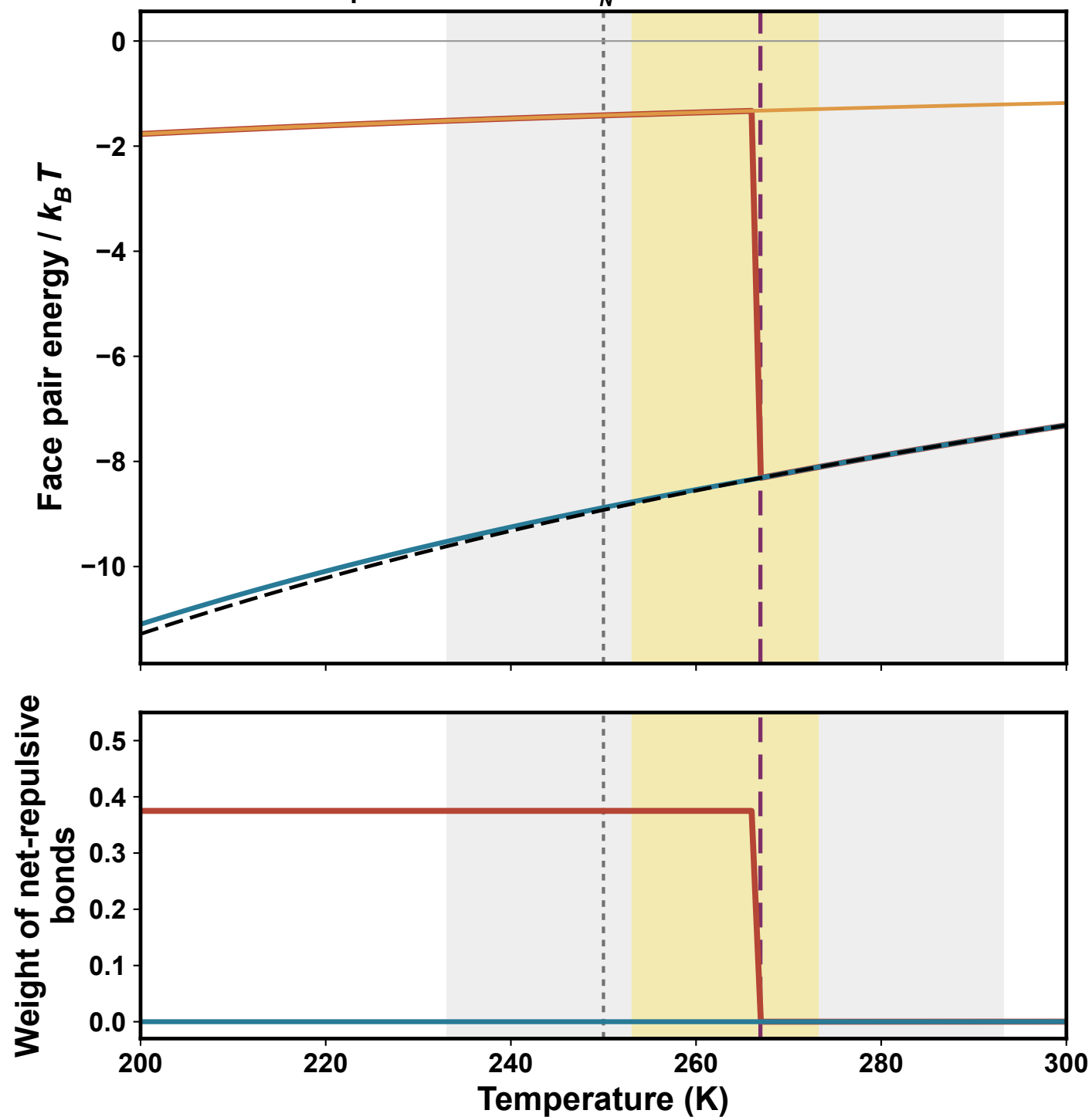


Cooling a fixed face-to-face contact from 300 K
dipoles freeze when τ_N exceeds the window



- Experimental range
- Transient aggregation (experiment)
- frozen random (inherited from dispersed)
- frozen ordered (inherited from bound)
- equilibrium reference (dipole never freezes)
- $E_{vdW}^{face,pair}$ (dipole averages to zero)
- freeze-in, $\tau_N = 20$ s (267 K)
- $T_B = 250$ K (100 s ZFC/FC)